

How much does the map of science depend on the embedding model?

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Abstract

Document embeddings make it possible to map scientific literature, but conclusions drawn from their geometry may depend on the model used. This study compares ten embedding models on a fixed corpus of 500,000 OpenAlex records from 26 fields and 2000–2024. Matched panels separate agreement between field centres, within fields and among nearest neighbours; further comparisons vary text input and representation settings. Models share broad structure, yet retain only 30.3% of the 25 nearest neighbours on average within field-period groups. Removing abstracts changes neighbours about as much as changing the model, whereas removing titles changes less. An exploratory analysis of all 325 field pairs identifies persistent opposite orderings across models in 262 pairs for angular spread and 221 for effective linear dimension. Requiring a relative difference above 5% reduces these counts to 96 and 177. Alternative measures preserve many oppositions, although centring and pooling matter. These results concern geometric properties, not thematic correctness. Repeating article selection can establish a stable result within a model while leaving its direction dependent on the representation. Science-mapping studies should therefore test the specific comparison they intend to interpret, alongside overall map agreement.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

Science maps help readers inspect relationships across a body of literature. They can bring neighbouring research areas into view, help locate related papers and support comparisons between fields. With text embeddings, these relationships begin with a model that converts each paper’s text into a vector, allowing papers to be compared through distances in a numerical space. Such representations have been used to map biomedical literature and examine the disciplinary structure of physics (González-Márquez et al., 2024; Constantino et al., 2025). But the model is part of how the map is made. How far would its conclusions change if the same papers were represented by another model?

The broader problem is familiar in science mapping. Gläser et al. (2017) asked how much a reconstructed thematic structure reflects scientific knowledge and how much it reflects the method used to identify it. In a comparison of text, citation and hybrid relations, Boyack and Klavans (2020) found substantial differences in cluster membership even when aggregate accuracy scores were similar. This matters when a map is used to support a specific claim: a summary of overall agreement may say little about whether two methods place the same papers together.

The present comparison follows that problem through a fixed corpus of 500,000 publications labelled English in OpenAlex, published in 2000–2024 and assigned to 26 broad Fields. Ten embedding models represent the same papers. The analyses examine relations between field centres, organisation within fields and the neighbours of individual articles. Separate matched subsets test changes to the input text and to how a model’s output is reduced to one vector per paper. A subsequent geometric analysis asks whether comparisons between fields retain their direction when both the model and the selected articles change. The aim is to identify which claims about the map survive these changes.

The results show shared broad relationships alongside changes in local neighbours and geometric comparisons between fields. Some models repeatedly place one field above another on a geometric measure, while other models repeatedly give the opposite order. The size of these differences matters, as does their sensitivity to processing choices. Repeating a comparison with different articles therefore addresses only part of its reliability. Agreement across representations provides useful evidence about what is shared, but does not establish which representation best captures the subject matter.

2 Background and research questions

2.1 Representation choices in science mapping

Comparisons of document embeddings already address several parts of this problem. Constantino et al. (2025) evaluated text and graph representations against the hierarchical classification of physics papers, including representations based on titles and abstracts. In biomedicine, González-Márquez et al. (2024) compared models and ways of reducing their outputs before selecting a representation for a large literature map. Their evaluation used the extent to which nearby papers shared field labels. Both studies make clear that a representation has to be assessed against a stated purpose and criterion.

Even within one model, the document representation is not fully specified by its name. Titles and abstracts supply different information. Models also differ in how much text they accept, while some require a choice of how to combine the representations of individual tokens into a single vector. This operation, usually called pooling, can use an average or a designated position in the output. Singh and Singh (2022) examined how scientific language models

respond to short queries and text perturbations, showing that changes in wording can alter retrieval behaviour. These choices belong in a comparison of the maps produced by the models.

There are also direct precedents for comparing the geometry of embeddings and their retrieved neighbours together. Caspari et al. (2024) used both types of comparison across retrieval datasets. Imel and Hafen (2025) examined local density and asymmetry across several document embeddings and disciplines in relation to citation rates. The present study builds on these comparisons by asking which broad, local and geometric conclusions persist across a fixed model panel, with matched articles and explicit controls for text, representation settings and the papers available as neighbours.

2.2 Agreement, stability and interpretation

Agreement and stability answer different questions here. Agreement concerns what two representations share when applied to the same articles. Stability concerns whether a specified result persists under a specified change, such as selecting different articles from the corpus. A field ordering may be stable within each model and still disagree between models. Thematic validity requires further evidence about how the represented relationships correspond to the research being described. In their study of invasion biology, Held and Velden (2022) showed why interpreting algorithmic groupings requires attention to the different relationships between papers that a mapping method can combine. Neither repetition nor model consensus supplies that interpretation on its own.

The term *shape* also needs an explicit object. Here it covers relationships in the embedding spaces, before any projection into two dimensions. Comparing the centres of fields describes a broad organisation; comparing articles within a field examines its internal structure. Neighbour agreement asks whether the same papers remain closest to a given article. Two further properties describe the angular spread of a field’s vectors and their effective linear dimension, which summarises how variation is distributed across directions. A field with greater geometric spread is not thereby more diverse in its topics, and effective linear dimension does not count those topics. The comparisons below retain these distinctions, so that a change in one property is not treated as evidence that every part of the map has changed.

2.3 Research questions and scope

Three questions organise the analysis:

RQ1. Which structural relationships are shared across embedding models, and how does agreement vary between field centres, articles within fields and local neighbours?

- RQ2.** How much do text input and pooling choices change geometry and neighbours relative to changing the embedding model?
- RQ3.** Which geometric comparisons between fields persist, reverse or remain unresolved across models and article selections, and how large are the differences?

These are descriptive questions about the selected corpus and models. The third question arose in an exploratory extension after the initial comparisons; the sequence of checks is documented in the methods and supplement. OpenAlex labels organise the comparisons and are retained with their known errors. They are not an independent reference for thematic correctness. The study consequently evaluates the persistence of conclusions within this design, without ranking models by their ability to recover a single correct map of science.

3 Data and comparative design

3.1 Corpus and analytical panels

The corpus contains 500,000 distinct OpenAlex work IDs, collected and cleaned by 15 September 2026. OpenAlex provides the metadata and texts (Priem et al., 2022); its core catalogue was restricted to articles, reviews and conference papers published in 2000–2024, labelled English, with a non-empty title, a reconstructible abstract of at least 50 words and a known primary Field. Retraction and paratext flags were excluded. Clear language and content failures were removed, while uncertain cases and possible duplicate versions were retained with flags. Each record’s primary-topic classification was frozen at extraction. These automatically assigned labels use text, citation and journal information (OpenAlex, 2026); they organise the analysis rather than supply an independent standard of correctness.

A general sample of 400,000 eligible records was drawn through seeded random API requests. An additional 100,000 records reinforced the smallest Field-period groups by repeatedly allocating places to the least populated group. Large groups were not reduced to the size of the smallest. The resulting 130 groups cover 26 Fields and five periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Every group contains at least 2,165 records. Global corpus comparisons use the general sample; group comparisons can use both cohorts. Equal-weight averages across groups describe a balanced comparison, not the proportions of world research.

Table 1 distinguishes the panels used below. The three-input panel contains 52,000 articles, equally divided across Field-period groups, and includes an earlier pilot of 26,000. A separate 52,000-article panel compares native model inputs with a shared literal text fragment that fits all tokenizers. Equal-size group comparisons and matched neighbourhood searches address different sources of variation. All comparisons preserve article identity between models. The extraction and cleaning rules, counts and retained flags are given in Supplement S1.

Table 1: Corpus and analytical panels.

Panel	Articles and representation	Comparison and scope
Original corpus	500,000 IDs: 400,000 base + 100,000 supplement. Ten models; title + abstract.	26 Fields, five periods. Global summaries use the base; the full corpus is not population-proportional.
Three text inputs	52,000 IDs; 400 per Field-period, 2,000 per Field. Title, abstract and both.	Same articles and candidates between models. Includes the 26,000-article pilot.
Common fragment	A different 52,000-ID panel; 400 per Field-period.	Paired native/shared content; model-specific tokenization. Distinct from the three-input panel.
Matched group size	256 articles per centre; 26 Fields, 217 Subfields, or 26 Subfield centres.	Within groups: 26 Fields and the same 183 Subfields at 128, 256 and 512 articles. Different objects from centres.
Paired searches	217 Subfields; 50 fixed queries each (10,850 total). Ten candidate selections.	256 candidates in each scope, including all 50 queries. The Field pool is conditioned on these queries.
Geometry / Field pairs	2,000 per Field; twenty halves of 1,000 and five selections of 2,000.	325 Field pairs and two properties; 26 saved conditions. Alternative metrics have different coverage.

Rows overlap and must not be added. Periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Each model comparison uses the same IDs. Query articles exclude themselves during neighbour search. The 1,300 technical test articles are not an additional scientific panel. Full filters and quotas: Table S1.

3.2 Models and text representations

Ten models cover scientific-document, general sentence and word-level biomedical/scientific representations (Table 2). This is a fixed panel chosen for its relevance to scholarly representation and contrasting training approaches, rather than a random sample of encoders or a ranking of the latest systems. Checkpoints, tokenizer revisions and text formats were frozen before inference. Each model encoded the same title and abstract in its customary format and with its native input limit. MiniLM’s 256-token limit was retained as primary; a separate 512-token variant tests that choice. Native truncation affected 49.58% of MiniLM inputs and 20.16% of MPNet inputs.

For BERT, SciBERT, BioBERT and PubMedBERT, the principal document vector is the mean of last-layer, non-padding token states, including special tokens. These models were not trained specifically for document similarity. Their first-position state (CLS, before the pooler) and final non-padding state (SEP) provide alternative readouts. The mean rule was specified before the primary comparisons and retained after the early sensitivities; it was not selected to maximise agreement. The other six models use their documented representation recipes. Title-only and abstract-only inputs use the same 52,000 IDs as the combined-input comparison. Native limits mean that this tests practical ways of using the models, without isolating architecture from text length.

Table 2: Embedding models and primary representation settings.

Model	Dim.	Tokens	Readout	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS, pre-pooler	(Gao et al., 2021)

MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens including special symbols. Mean: last-layer average over non-padding positions, including special symbols. CLS: first-position state before the pooler; SPECTER2 adds its proximity adapter. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact check-point training. Readout variants are not additional models. Full revisions, formats and truncation: Table S2.

3.3 Objects and measures

All measurements use the original embedding dimensions, without a two-dimensional projection. Article vectors are first scaled to unit length. Shape agreement is measured primarily by corrected linear centred kernel alignment (CKA), which compares relationships among the same identified articles (Kornblith et al., 2019; Murphy et al., 2024). Higher values indicate greater agreement. The finite-sample correction reduces a known source of inflated alignment; it does not make the complete ratio an exactly unbiased estimator. Procrustes

alignment and correlations between distance ranks provide checks with different sensitivities. Their definitions and the correction formula appear in Supplement S2. These related measures are not independent votes about validity (Williams et al., 2021; Williams, 2024).

A group centre is the mean of its unit-length article vectors. Comparisons between centres describe broad organisation, while comparisons within groups retain article-level variation. Random groupings preserve group sizes and publication periods, using the same allocation across models, to assess how much centre agreement accompanies averaging without the observed labels.

Neighbour agreement is the fraction of the k closest articles shared by two models. The main choice is $k = 25$, with 10 and 50 as checks. Searches use cosine similarity and exclude the query article itself. The candidate pool is stated for each result: all articles in a Field-period group, an equal-size subset, a matched Field/Subfield pool, or the general corpus. An exact blockwise calculation avoids storing every pairwise distance. Independent full-sort checks verify saved neighbour lists. Neither overlap nor its simple chance-adjusted form removes all effects of candidate composition or size.

Two additional properties describe each Field geometrically. *Angular spread* is the median angle between distinct article vectors. *Effective linear dimension* is the participation ratio of the centred covariance matrix C , $PR = (\text{tr } C)^2 / \text{tr}(C^2)$. It is larger when variation is distributed across more directions. Angular quantiles, angles to the centre, entropy-based effective rank and the number of directions containing 80% of variance serve as alternatives. The latter two use the same covariance spectrum as PR; they do not provide independent thematic validation. Graph-connection measures were also tested, but their synthetic counterexamples prevented an interpretation as general fragmentation (Supplement S6).

3.4 Matched comparisons and stability

The strict Field/Subfield neighbour comparison fixes 50 query articles in each of 217 Subfields. Both searches contain 256 candidates, including those same 50 queries, and identical period counts. Ten selections vary the remaining 206 candidates within the Subfield or its parent Field. Thus, the broader search is conditioned on the queries; it is not an unrestricted draw of 256 Field articles. The design covers 10,850 queries and holds search opportunity constant while varying its scope.

For geometric Field comparisons, every unordered pair of the 26 Fields is evaluated: 325 pairs for each property. A model gives a persistent direction only if that direction holds across the primary 2,000 articles per Field, twenty half samples of 1,000 and five additional selections of 2,000 from the same corpus. All-ten agreement requires the same persistent direction in every model. Opposition requires at least two fixed models with persistent opposite directions. Remaining pairs are unresolved. This does not mean equality. Relative

magnitude is $2(B - A)/(A + B)$, where A and B are the two Field values; minima of 0, 1, 5 and 10% test sensitivity to small differences.

Repeated selections measure sensitivity within the frozen corpus. They are overlapping subsets, not independent population samples or confidence intervals. Operational screens retain cases with excessive median shifts or selection ranges; their rules and separate denominators are reported in the supplement. Model pairs share models, and Field pairs share Fields, so pair counts are treated descriptively. No significance test treats those pairs as independent observations.

3.5 Sequence of checks

The analysis developed in stages. The primary shape and neighbour comparisons preceded the expanded input panel, matched Subfield searches and atlas of concrete cases. Morphology was an exploratory extension, followed by the all-pairs summary after its initial examples were known. Rules were recorded locally before the corresponding new calculations; this is not an external preregistration of the whole study. Some controls, including random-group centres and a fixed-neighbour-fraction check, were added after earlier results. Supplement S2 records that sequence. Time, discipline composition, model traits, source errors and processing sensitivities are retained in Supplements S3–S7, including checks that limit the interpretation.

4 Results

4.1 Shared broad structure and within-field variation

Models agree strongly about the arrangement of group centres. With 256 articles contributing to each centre, mean corrected CKA is 0.939 for the 26 Fields and 0.911 for 217 Subfields (Figure 1A). Restricting the latter to 26 Subfield centres gives 0.902. Random groups with matched sizes and dates also show positive agreement, around 0.63–0.64, but do not recover the observed level. Averaging therefore contributes to the comparison, yet is insufficient to account for the agreement associated with these labelled groups.

Within groups, mean CKA remains around 0.63 for Fields and 0.64 for the same 183 Subfields eligible at all three sizes (Figure 1B). Increasing the number of articles from 128 to 512 changes these means little. Narrower labels do not produce a general reduction in agreement under this comparison. Centres and their constituent articles are different objects, however: the difference between the panels is not an isolated causal effect of scale.

The alternative measures broadly support the ordering of model-pair agreements. Across the original 130 Field-period groups, the mean rank correlation with corrected CKA is 0.891

for Procrustes and 0.881 for distance ranks. Their agreement is incomplete, and the original selection screen retains 50 alerts among 5,850 model-pair/group comparisons (Table S4). These qualifications concern particular comparisons; an overall mean does not erase them.

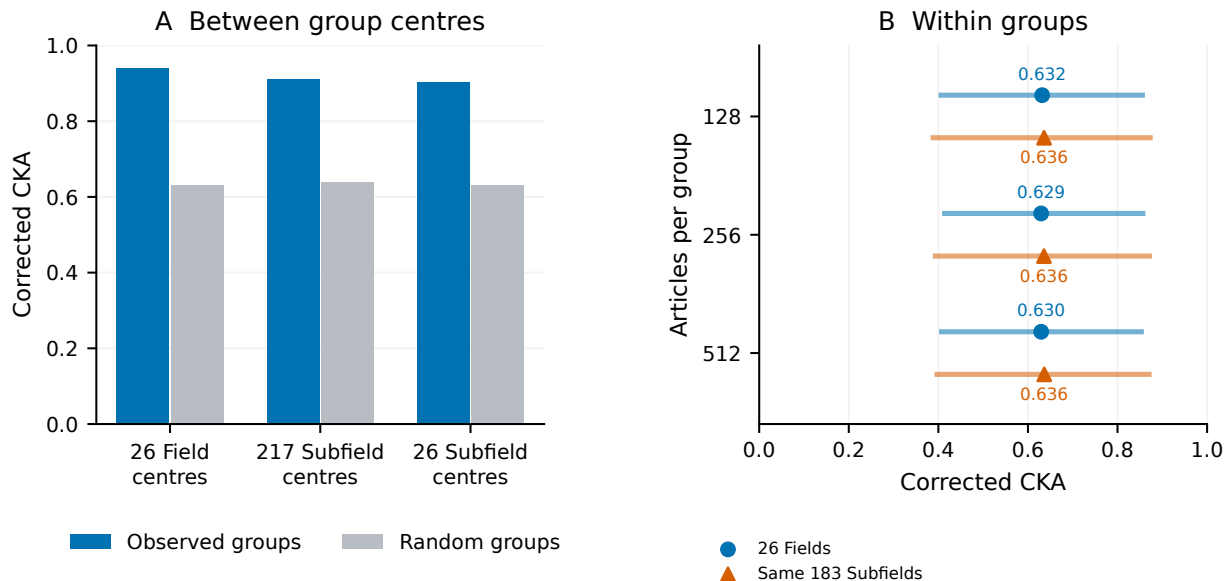


Figure 1: Agreement between group centres and within groups. (A) Mean corrected CKA across the fixed model pairs and saved centre comparisons; each centre contains 256 articles. Random groups preserve sizes and dates. The 26-Subfield condition controls the number of centres. (B) Within-group agreement at three article counts, for 26 Fields and the same 183 eligible Subfields. Points and labels show means; lines span the 2.5th–97.5th percentiles across group–model-pair comparisons (1,170 for Fields; 8,235 for Subfields). These are descriptive spreads, not confidence intervals; model pairs are not independent. All CKA axes span 0–1. The panels compare different objects, so their difference is not an estimated causal effect.

4.2 Neighbour agreement under matched comparisons

Across the full Field-period searches, two models retain an average of 30.3% of the 25 nearest neighbours: about 7.6 articles out of 25. This average gives each group and model pair equal weight. Restricting every group to 2,048 common candidates raises overlap to 31.9%. In the separate search across the 400,000-article general corpus, 13,000 balanced queries retain 17.5%. The last result changes both the scope and the number of available candidates, so it cannot by itself establish weaker agreement across disciplinary boundaries.

The matched Field/Subfield experiment addresses that ambiguity directly (Figure 2). With the same queries, dates and 256-candidate size, Subfield agreement is lower in 127 of 217 cases and higher in 90. The mean difference is small, at -1.35 percentage points. For 50

neighbours, the split becomes 109 lower and 108 higher. A narrower search can preserve more neighbours or fewer, depending on the group and the chosen neighbourhood size. There is no universal loss of agreement from Field to Subfield.

Shape and neighbour agreement nevertheless move together. Across the 45 model-pair means in the original analysis, their descriptive rank correlation is 0.920. Reporting both is useful because they describe different consequences of representation choice, rather than because they form unrelated dimensions of reliability. CKA compares distributed relationships; a neighbour list concerns which specific articles a reader would encounter.

The atlas makes those local differences inspectable. Its Subfield examples are selected by a recorded persistence rule across 27 combinations of size, readout and neighbour count. Article-level examples then show links that persist across models alongside links whose inclusion varies sharply (Supplement S4). Some extreme cases expose source problems: a history book appears under Physics, an infection-related record under Political Science, and an announcement remains in the corpus. Comparison with the original records confirms that these are source or classification issues, rather than mismatched embedding rows. The examples are exploratory and deliberately selected; they estimate neither the prevalence of such errors nor the accuracy of the links.

4.3 Text input and representation choices

Removing the abstract can change a representation about as much as changing the encoder. On the same 52,000 articles, the mean shape change, measured as one minus corrected CKA, is 0.370 when changing model and 0.366 when retaining the model but using only the title. The corresponding fractions of neighbours replaced are 68.8% and 70.7% (Figure 3). These close averages do not establish equivalence. They show that a seemingly simple input choice can be comparable in scale to the model choice in this design.

Removing only the title changes less. Abstract-only inputs have a mean shape change of 0.027, yet replace 22.1% of the 25 neighbours. A small change in overall geometry can therefore still alter the papers returned locally. The model-specific results also vary substantially: title-only neighbour replacement ranges from 51.7% for MPNet to 84.2% for SciBERT (Table S5). A single pooled mean would conceal those differences.

Pooling affects the comparison itself. Under the principal mean readout, title removal from the abstract is relatively modest, while discarding the abstract slightly exceeds model change in mean neighbour replacement. Switching the four word-level BERTs to CLS or SEP reverses that latter average ordering. Those scenarios also change the cross-model reference, because four representations move together. They do not identify a pure effect of pooling in isolation, and offer no reason to select whichever readout yields the most agreement.

The larger input panel reduces sensitivity to article selection. Using 100 comparable selections at each size, the shape-contrast screen drops from 72 alerts among 520 model–Field–

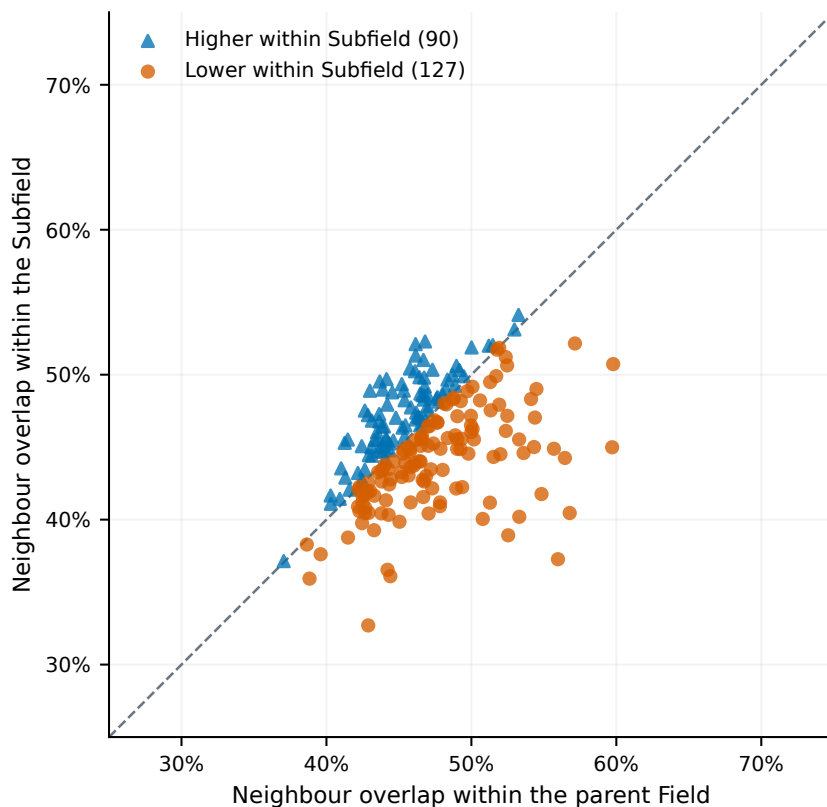


Figure 2: Neighbour agreement in matched Field and Subfield searches. Each point is one of 217 Subfields, averaged over model pairs and ten candidate selections. Both scopes contain the same 50 fixed queries within 256 candidates and matched dates; self is excluded. The broader Field pool is conditioned on these queries. Points above the diagonal have higher overlap in the narrower search; those below have lower overlap. Dispersion describes group differences, not confidence intervals.

input contrasts at 26,000 articles to five at 52,000. The earlier 20-selection check gives 31 and four, respectively (Figure S1). Both records are retained. Passing this screen does not mean that a contrast differs from zero, nor does it certify all neighbour results or readout variants. The common-fragment and MiniLM-length checks further distinguish unequal text exposure from a coding error: the original MiniLM limit was its documented setting, not an implementation fault.

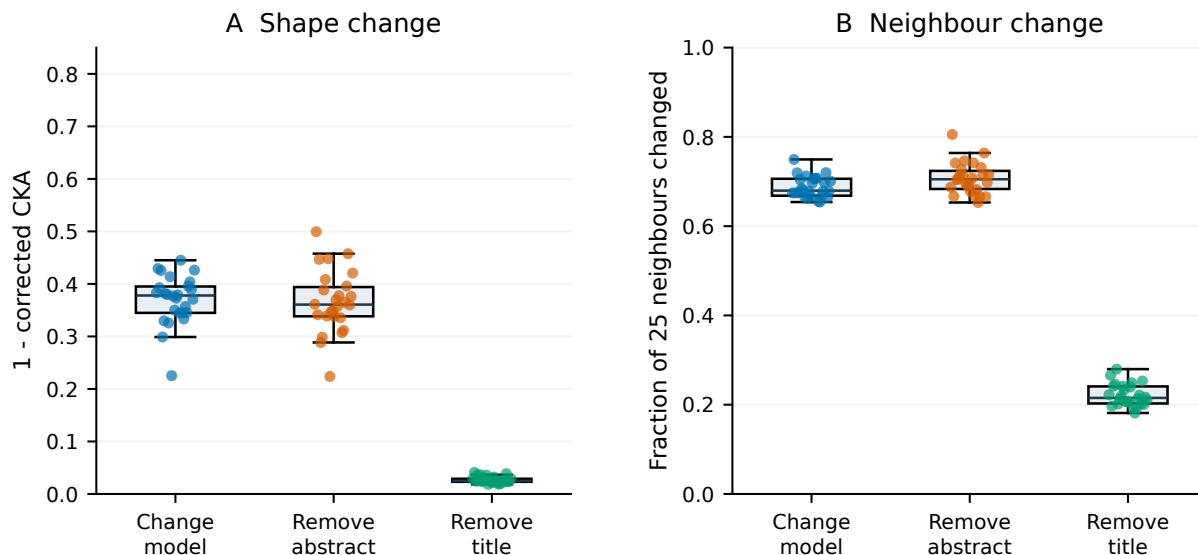


Figure 3: Changes associated with model and text input. The same 52,000 IDs provide 2,000 candidates per Field. The reference input is title plus abstract: “remove abstract” leaves the title, and “remove title” leaves the abstract. (A) One minus corrected CKA. (B) Fraction of the 25 neighbours replaced. Each point is a Field mean over the fixed model panel. Boxes show medians and interquartile ranges, with 1.5-IQR whiskers; all Field points are displayed. These are not confidence intervals. Native text limits are retained.

4.4 Persistence and reversal of geometric field comparisons

The geometric extension asks a more specific question than whether two maps resemble each other: would changing the model reverse a comparison between two Fields? Article selection alone usually preserves the measured spread and effective linear dimension. Angular spread produces no alerts in the 260 model–Field cases under either the half-sample or equal-size selection screen. Effective dimension retains four half-sample alerts, although none appears in the five equal-size selections (Table S12). Those four cases remain part of the reported evidence.

Stability within a model does not ensure agreement between models. Among all 325 Field pairs, angular spread has a persistent common direction in all ten models for 42 pairs, persistent opposition for 262 and an unresolved outcome for 21. Effective linear dimension has corresponding counts of 54, 221 and 50 (Figure 4). Opposition means that at least two particular models repeatedly support opposite directions across all 26 saved selection conditions. It does not require every model to disagree, and unresolved pairs can contain partial agreement.

Many angular differences are small. Requiring each opposing direction to exceed a symmetric relative difference of 5% reduces angular opposition to 96 pairs; at 10%, 19 remain. Effective linear dimension retains 177 and 126 oppositions at those cutoffs. Thus, the direction-only totals should not be read as counts of equally large disagreements. The minimum is an explicit sensitivity choice, not a universal definition of a meaningful scientific difference. Figure 4 retains all 325 pairs in each row so that the growing unresolved category remains visible.

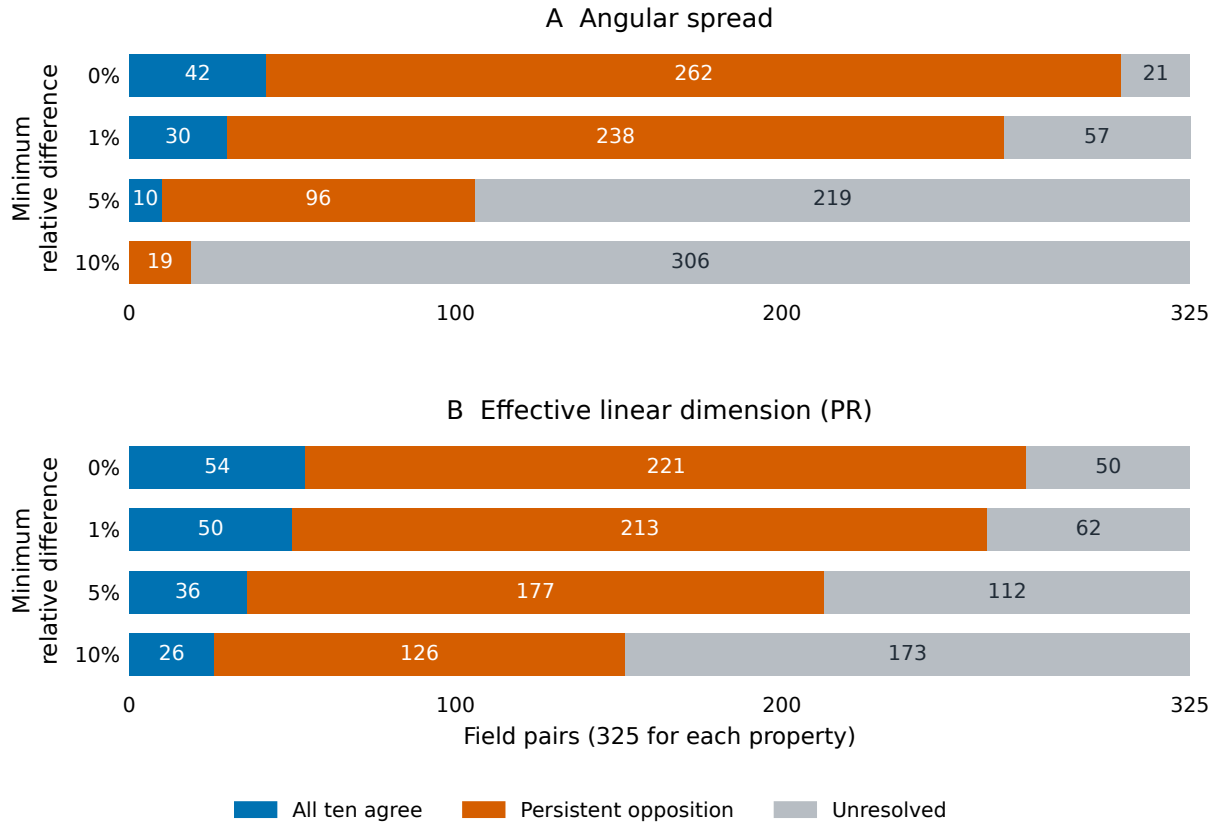


Figure 4: Persistence, opposition and magnitude of Field comparisons. Each row contains all 325 Field pairs for one property and ten models. A model direction must persist across the primary selection, twenty half samples and five additional equal-size selections from the same corpus. All-ten agreement requires a common persistent direction; opposition requires at least two fixed models with persistent opposite directions. All remaining pairs are unresolved. Magnitude uses $2(B - A)/(A + B)$. The four minima are operational cutoffs, not significance or accuracy thresholds. The 26 conditions overlap; point controls are reported separately.

Alternative measures preserve much of the pattern. Requiring the same opposing models to keep their directions across the angular alternatives retains 225 of the original 262 oppositions. The spectral alternatives retain 196 of 221. Their coverage differs: angular alternatives have all 26 selection conditions, whereas entropy-based rank and the 80%-variance count are available for three. These alternatives also share much of their mathematical construction.

Their agreement supports a geometric description without turning it into a measure of topic diversity.

The result is not confined to the four word-level BERTs. Restricting the panel to SPECTER, SPECTER2, SciNCL, MPNet, MiniLM and SimCSE still yields 231 angular and 160 effective-dimension oppositions at zero minimum. A smaller panel has fewer opportunities to contain opposing models, so the change in counts does not isolate an effect of training objective. All model omissions and the complete pair classifications accompany Tables S14–S15.

Processing choices set a further boundary. After subtracting the global centre and renormalising, the same original opposing model pair and directions survive for 145 of 262 angular oppositions and 218 of 221 effective-dimension oppositions in the matched point check. Angular ordering is therefore more dependent on that choice. Pooling changes can also reverse comparisons (Table S16). These are point sensitivities with matched references, not another set of 26-repeat confirmations. The retained conclusions concern explicitly defined representations, rather than properties expected to survive any transformation.

Connection measures were less defensible. Elongated clouds and isolated extreme points could resemble fragmentation under some rules, while changes in sample size or the number of neighbours changed the result substantially. The equal-size selection screen alone retained 51 connection alerts among 260 cases. Consequently, connection is reported with its failed construct checks in Supplement S6 and is not used to claim that one Field contains more separate thematic groups.

5 Discussion

5.1 From map agreement to the persistence of a claim

A model can reproduce the broad organisation of a corpus while changing a particular comparison made within it. The centre results show substantial shared structure, well above the random-group reference. The Field-pair results show why that alone is insufficient for interpreting geometric rankings. Even after repeatedly changing the articles, one model can consistently give the opposite ordering from another. The useful robustness question is therefore attached to the intended claim: which relationship, neighbour set or Field comparison needs to survive representation choice?

This extends a comparative problem already recognised in science mapping (Gläser et al., 2017; Boyack and Klavans, 2020). Evaluation against a classification, as in Constantino et al. (2025), addresses how well a representation serves a particular external criterion. Comparing geometric alignment and retrieval, as in Caspari et al. (2024), establishes which relationships the models share. The contribution here is to connect those levels to the persistence and

magnitude of specific Field comparisons on matched articles. It does not rest on introducing agreement measures or on claiming that earlier studies ignored representation choice.

Angular spread and effective linear dimension are useful precisely when their meaning remains narrow. An angle measures geometric separation; PR measures how variance occupies linear directions. Interpreting either as interdisciplinarity or semantic diversity would require additional evidence about the underlying content. Similar cautions apply to claims about embedding isotropy (Rudman et al., 2022). The rejected fragmentation interpretation illustrates the cost of skipping that step: a measure may distinguish clouds reliably while failing to identify the substantive property suggested by its name.

5.2 Implications for comparative science mapping

A practical analysis should fix the claim, input and comparison opportunities before choosing a robustness summary. For document retrieval, candidate pools and neighbour count are part of the question. For Field geometry, the representation rule, article balance and minimum difference matter. The input experiment also makes a straightforward reporting requirement: naming the encoder is insufficient. The title/abstract choice, token limit and pooling rule should be recorded alongside it.

Additional comparisons caution against simple explanations of agreement. Medicine has lower shape agreement than Energy, but is not consistently lowest in neighbour agreement. Balancing Subfield composition and excluding BioBERT and PubMedBERT leaves the Medicine–Energy difference; neither factor supplies a unique cause (Supplement S5). Apparent model families also change with readout. The ten encoders differ in several linked ways, so their similarities cannot isolate the influence of training corpus or objective.

Time provides another example of why the comparison design matters. Mean shape agreement rises slightly between the first and last periods. Mean neighbour agreement falls by 0.87 percentage points with all candidates but rises by 0.91 points with 2,048 candidates per group. The sign therefore depends on search opportunity. These models represent older and newer publications using their current parameters; they do not reconstruct the knowledge available when those publications appeared. The result offers no causal evidence that science itself has converged.

5.3 Limits and next validation

The corpus covers selected OpenAlex records with usable abstracts and predominantly English text. The reinforcement sample, balanced panels and equal-weight summaries serve comparisons across Fields; they are not estimates of the worldwide distribution of research. Classification and content errors remain, as the atlas demonstrates. The quality checks remove specified problems rather than establish that every article is correctly labelled. Local

selections and frozen checkpoints make the comparison auditable, but repeated sampling from this corpus cannot remove coverage bias.

The fixed model panel is similarly bounded. Its members share architectures, training sources and objectives, and several are not specialised similarity encoders. Sensitivities over six similarity-oriented models and alternative readouts narrow some concerns without making the panel representative of all models. Persistent opposition establishes model dependence within that panel; it does not tell us which direction is substantively correct.

Finally, the geometry extension and its all-pairs summary followed earlier results. The recorded sequence and complete tables make those choices inspectable, but do not convert exploration into confirmation. A focused next study could test one consequential comparison against independently assessed article relationships or a discipline-specific question. That would address the thematic interpretation left open here, rather than simply add more encoders to the same agreement calculation.

6 Conclusion

The ten models share broad structure, but that agreement does not guarantee that they preserve the same neighbours or the same ordering of Fields on a geometric property. Input choice can be as consequential as model choice, and a comparison can remain stable under repeated article selection while reversing across encoders. Testing the intended scientific claim across representations is therefore more informative than reporting a single overall similarity score. Angular spread and effective linear dimension support such a test when their definitions, effect sizes and processing sensitivities remain explicit. They describe the geometry produced by the models; interpreting that geometry as a property of scientific knowledge requires further validation.

Declarations

Author contributions. Alejandro Treny Ortega conceived the project and directed the research and manuscript preparation as an independent researcher. Computational and editorial assistance is disclosed below.

Competing interests. The author declares no competing interests.

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Data and code availability. The project repository is <https://github.com/aleetreney/Shape-of-Science-Robustness>. This draft is accompanied by the tables used for its figures and the editable manuscript sources. The frozen corpus, vectors and complete

local execution records have not been deposited in a persistent public archive. Public code alone does not provide full reproduction of the present analysis. The access inventory and remaining release requirements are stated in Supplement S8; no data deposit or persistent identifier is claimed here.

Use of AI-assisted tools. OpenAI Codex assisted with literature searches, software development and analysis, visualisation, and manuscript drafting and editing. The author is responsible for the content.

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